

Web Designing & Publishing



Chapter : 8

Web Publishing and Browsing

Web Publishing

- Web publishing is the process of publishing original content on the Internet.
- The process includes building and uploading websites, updating the associated webpages, and posting content to these webpages online.
- The content meant for web publishing can include text, videos, digital images, artwork, and other forms of media.
- Publishers must possess a web server, a web publishing software, and an Internet connection to carry out web publishing.
- Web publishing is also known as online publishing



Overview of World Wide Web

- The **world wide web(WWW)** is the most popular and promising method of accessing the internet.
- The WWW is an information system designed specifically for posting useful information which can be used with ease and with document interchange in mind.
- **HTML (Hypertext Markup Language)** is developed for this purpose.
- HTML was based on a subset of **Standard Generalized Markup Language (SGML)**.
- To transfer HTML document to remote sites, **HTTP (Hypertext Transfer Protocol)** was derived.



SGML (Standard Generalized Markup Language)

- SGML (Standard Generalized Markup Language) is a standard for how to specify a document markup language or tag set. Such a specification is itself a **Document Type Definition (DTD)**.
- SGML is not in itself a document language, but a description of how to specify one. It is metadata.
- SGML is based on the idea that documents have structural and other semantic elements that can be described without reference to how such elements should be displayed.
- The actual display of such a document may vary, depending on the output medium and style preferences. Some advantages of documents based on SGML are:



SGML Conti..

1. They can be created by thinking in terms of document structure rather than appearance characteristics (which may change over time).
2. They will be more portable because an SGML compiler can interpret any document by reference to its document type definition (DTD).
3. Documents originally intended for the print medium can easily be re-adapted for other media, such as the computer display screen.



Web hosting

- Web hosting is a service of providing online space for storage of web pages. These web pages are made available via World Wide Web. The companies which offer website hosting are known as **Web hosts**.
- The servers on which web site is hosted remain switched on 24 x7. These servers are run by web hosting companies. Each server has its own IP address. Since IP addresses are difficult to remember therefore, webmaster points their domain name to the IP address of the server their website is stored on.
- It is not possible to host your website on your local computer, to do so you would have to leave your computer on 24 hours a day. This is not practical and cheaper as well. This is where web hosting companies comes in.



Types of Hosting

- Shared Hosting
- Virtual Private Server (VPS)
- Dedicated Server



Shared Hosting

In shared hosting, the hosting company puts thousand of website on the same physical server. Each customer has their own allocation of physical web space and a set of bandwidth limit. As all websites share same physical memory, MYSQL server and Apache server, one website on the server experiencing high traffic load will affect performance of all websites on the server.



Virtual Private Server (VPS)

- It is also known as Virtual Dedicated Server.
- It is a server which is partitioned into smaller servers.
- In this customer is given their own partition, which is installed with its own operating system.
- Unlike shared hosting, VPS doesn't share memory or processor time rather it allocates certain amount of memory and CPU to use which means that any problem on a VPS partition on the same drive will not affect other VPS customers



Dedicated Server

In this kind of hosting, single dedicated server is setup for just one customer. It is commonly used by the businesses that need the power, control and security that a dedicated server offers.



Basic feature of web hosting

- Disk Space
- Monthly Traffic
- Bandwidth
- Connection speed
- Email Support
- Programming Service
- Uptime
- Backup & FTP
- Control Panel



Components of web publishing

- Domain name
- Web hosting (Web Space)
- Web design and development
- Promotion
- Maintenance



Documents Interchange Standards

XML is an example of a document interchange standard that can be used to create and share revisable documents

- XML stands for **eXtensible Markup Language**
- XML is a markup language much like HTML
- XML was designed to store and transport data
- XML was designed to be self-descriptive
- XML is a W3C Recommendation



The Difference Between XML and HTML

XML and HTML were designed with different goals:

- XML was designed to carry data - with focus on what data is
- HTML was designed to display data - with focus on how data looks
- XML tags are not predefined like HTML tags are



Example of XML

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<bookstore>
```

```
<book category="cooking">
  <title lang="en">Everyday Italian</title>
  <author>Giada De Laurentiis</author>
  <year>2005</year>
  <price>30.00</price>
</book>
```

```
<book category="children">
  <title lang="en">Harry Potter</title>
  <author>J K. Rowling</author>
  <year>2005</year>
  <price>29.99</price>
</book>
```

```
<book category="web">
  <title lang="en">XQuery Kick Start</title>
```

```
<author>James McGovern</author>
<author>Per Bothner</author>
<author>Kurt Cagle</author>
<author>James Linn</author>
<author>Vaidyanathan Nagarajan</author>
<year>2003</year>
<price>49.99</price>
</book>
```

```
<book category="web" cover="paperback">
  <title lang="en">Learning XML</title>
  <author>Erik T. Ray</author>
  <year>2003</year>
  <price>39.95</price>
</book>
```

```
</bookstore>
```



Web Page Design Considerations and Principles

- Easy Navigation
- Responsive design
- Same Color Scheme (Consistency)
- Comfortable UI
- Contents meet user goals for visiting websites
- Load time
- use of Image



Search and Meta Search Engines

A metasearch engine (or search aggregator) is an online information retrieval tool that uses the data of a web search engine to produce its own results. Metasearch engines take input from a user and immediately query search engines for results. Sufficient data is gathered, ranked, and presented to the users.

Examples of metasearch engines include **Skyscanner** and **Kayak.com**, which aggregate search results of online travel agencies and provider websites and Excite, which aggregates results from internet search engines.



Need for Metasearch Engines

The Metasearch Engines were developed to cover the entire web, unlike most search engines. Individual search Engines try to spam people to enhance their page rankings. This is an illegitimate way of promoting. The individual Search engines are unable to find results from other search engines. This is when the Metasearch Engine comes handy. This also supports multiple formats, unlike individual engines. The meta search engines seem effortless.



The Metasearch Engine Architecture

User Interface: The user interface of the Metasearch Engine is similar to the look and feel of the individual search engines like Google and Yahoo. It even has options to search on the basis of types and categories and also of which search engines it must use to write back the result.

Dispatcher: The dispatcher has the work of query generation.

Display: The display uses the queries to write back the results on the screen. It uses methods such as page ranks, parsing techniques, cluster formation, and stitching to give the desired result.

Personalization: The personalization in other words is being user-specific. This involves comparing the results with each other.



WWW (World Wide Web)

World Wide Web, which is also known as a Web, is a collection of websites or web pages stored in web servers and connected to local computers through the internet. These websites contain text pages, digital images, audios, videos, etc. Users can access the content of these sites from any part of the world over the internet using their devices such as computers, laptops, cell phones, etc. The WWW, along with internet, enables the retrieval and display of text and media to your device.



THANK YOU

Address: G. R. Complex Preetam Nagar, Dhoomanganj
Prayagraj Uttar Pradesh 211011

Mobile No.: 9598948810, 8874588766, 9532878816

Website : www.infomaxacademy.com

